

AUDIT_IMPLEMENTATION_PLAN

HelixCode Audit Implementation Plan

Created: 2026-01-09 **Status:** IN PROGRESS **Last Updated:** 2026-01-09

This document tracks the implementation of fixes identified in the Comprehensive Audit Report.

Progress Tracking

Overall Status

Phase	Status	Progress	Blockers
P0 Critical - Mock Data	NOT STARTED	0%	None
P0 Critical - Dependencies	NOT STARTED	0%	None
P0 Critical - Security	NOT STARTED	0%	None
P1 High - Stubs	NOT STARTED	0%	Depends on P0
P1 High - Coverage	NOT STARTED	0%	Depends on P0
P2 Medium - Apps	NOT STARTED	0%	Depends on P1
P2 Medium - Docs	NOT STARTED	0%	None
P3 Low - Remaining	NOT STARTED	0%	Depends on P2

Phase 0: CRITICAL Fixes

Task P0-1: Remove Mock Data from handlers.go

File: internal/server/handlers.go **Priority:** P0 - CRITICAL **Estimated Effort:** 8 hours **Assigned To:** TBD

Subtasks



- P0-1.1** Fix createTask() (lines 392-405)
- Replace placeholder ID with proper error response
 - Return HTTP 503 when database unavailable

- Add test for this behavior



P0-1.2 Fix `getTask()` (lines 429-442)

- Remove “Sample Task” hardcoded data
- Return HTTP 503 or 404 appropriately
- Add test for database unavailable scenario



P0-1.3 Fix `updateTask()` (lines 507-521)

- Remove placeholder data return
- Return proper error when database unavailable
- Add test coverage



P0-1.4 Fix `updateProject()` (lines 287-302)

- Remove hardcoded “/path/to/project”
- Implement actual project update logic
- Add test coverage



P0-1.5 Fix `deleteProject()` (lines 305-311)

- Implement actual deletion logic
- Return error if delete fails
- Add test coverage



P0-1.6 Fix `getWorker()` (lines 593-605)

- Remove hardcoded “localhost”
- Return proper error when worker not found
- Add test coverage

Verification: - [] All tests pass - [] No placeholder data in responses - [] Coverage increased

Task P0-2: Fix Authentication Bypass

Files: - `internal/server/handlers.go` - `internal/project/manager_db.go`

Priority: P0 - CRITICAL **Estimated Effort:** 4 hours **Assigned To:** TBD

Subtasks



P0-2.1 Remove "default-user" fallback in `listProjects()` (`handlers.go:18-22`)

- Return HTTP 401 Unauthorized when `user_id` missing
- Add proper authentication middleware check
- Add test for unauthorized access

**P0-2.2** Fix `CreateProject()` (`manager_db.go:134-135`)

- Remove hardcoded “default-user”
- Require user ID parameter
- Update all callers
- Add test coverage

Verification: - [] No hardcoded user IDs - [] Proper 401 responses for unauthenticated requests - [] Tests cover auth scenarios

Task P0-3: Fix Confirmation/Permission Bypass

Files: - `internal/tools/confirmation/prompter.go` - `internal/workflow/autonomy/permission.go`

Priority: P0 - CRITICAL **Estimated Effort:** 6 hours **Assigned To:** TBD

Subtasks**P0-3.1** Fix auto-allow in `prompter` (`prompter.go:179-185`)

- Remove automatic `ChoiceAllow` return
- Require actual input source
- Return error when no input available
- Add test for this behavior

**P0-3.2** Fix mock confirmation (`permission.go:232-242`)

- Implement actual confirmation flow
- Block until user responds or timeout
- Add test coverage

Verification: - [] No automatic permission grants - [] Confirmation required for sensitive actions - [] Tests verify blocking behavior

Task P0-4: Security Dependency Updates

File: `go.mod` **Priority:** P0 - CRITICAL **Estimated Effort:** 4 hours **Assigned To:** TBD

Subtasks**P0-4.1** Remove deprecated Redis v8

go mod edit -droprequire github.com/go-redis/redis/v8

- Update all imports to use v9
- Verify no v8 references remain



P0-4.2 Update Redis v9

go get github.com/redis/go-redis/v9@v9.17.2

- Test Redis functionality



P0-4.3 Update golang.org/x/crypto

go get golang.org/x/crypto@v0.46.0

- Run security tests



P0-4.4 Replace nfnt/resize

- Find modern image resize library
- Update all usages
- Add deprecation tests



P0-4.5 Replace golang.org/x/freetype

- Find modern font library
- Update all usages



P0-4.6 Run go mod tidy

- Verify build passes
- Run full test suite

Verification: - [] go mod graph shows no deprecated deps - [] All tests pass - []
Security scan clean

Task P0-5: Implement Network Isolation

File: internal/tools/shell/sandbox.go **Priority:** P0 - CRITICAL **Estimated Effort:** 8 hours **Assigned To:** TBD

Subtasks

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P0-5.1 Design network isolation approach

- Research Linux network namespaces
- Document implementation strategy

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P0-5.2 Implement for Linux (lines 190-196)

- Create network namespace
- Apply firewall rules
- Test isolation

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P0-5.3 Add fallback for other platforms

- Return error on unsupported platforms
- Document limitations

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P0-5.4 Add comprehensive tests

- Test network is isolated
- Test fallback behavior

Verification: - [] Network isolated in sandbox - [] Tests verify isolation - []
Documentation updated

Phase 1: HIGH Priority Fixes

Task P1-1: Implement AI Provider Wrappers

File: internal/providers/ai_integration.go **Priority:** P1 - HIGH **Estimated Effort:** 16 hours **Assigned To:** TBD

Subtasks

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P1-1.1 Implement OpenAI wrapper (line 1326)

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P1-1.2 Implement Anthropic wrapper (line 1329)

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P1-1.3 Implement Cohere wrapper (line 1332)

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P1-1.4 Implement HuggingFace wrapper (line 1335)

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P1-1.5 Implement Mistral wrapper (line 1338)

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- ☐ **P1-1.6** Implement Gemini wrapper (line 1341)
- ☐ **P1-1.7** Implement Gemma wrapper (line 1344)
- ☐ **P1-1.8** Implement LlamaIndex wrapper (line 1347)
- ☐ **P1-1.9** Implement MemGPT wrapper (line 1350)
- ☐ **P1-1.10** Implement CrewAI wrapper (line 1353)
- ☐ **P1-1.11** Implement CharacterAI wrapper (line 1356)
- ☐ **P1-1.12** Implement Repika wrapper (line 1359)
- ☐ **P1-1.13** Implement Anima wrapper (line 1362)
- ☐ **P1-1.14** Add tests for all wrappers

Verification: - [] No NotImplementedProvider returns - [] All wrappers functional - [] Tests pass

Task P1-2: Complete Zep Provider

File: internal/memory/providers/zep_provider.go **Priority:** P1 - HIGH
Estimated Effort: 6 hours **Assigned To:** TBD

Subtasks

- ☐ **P1-2.1** Implement Retrieve () (line 207)
 - Research Zep API for ID retrieval
 - Implement proper retrieval
 - Add tests
- ☐ **P1-2.2** Implement Update () (line 213)
 - Research Zep API for updates
 - Implement proper update
 - Add tests
- ☐ **P1-2.3** Implement Delete () (line 221)
 - Research Zep API for deletion
 - Implement proper deletion
 - Add tests

Verification: - [] CRUD operations work - [] No stub warnings logged - [] Tests pass

Task P1-3: Implement Tree-Sitter Parser

File: internal/tools/mapping/treesitter.go **Priority:** P1 - HIGH **Estimated Effort:** 8 hours **Assigned To:** TBD

Subtasks

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P1-3.1 Add tree-sitter dependency

```
go get github.com/smacker/go-tree-sitter
```

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P1-3.2 Implement Go parser

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P1-3.3 Implement Python parser

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P1-3.4 Implement JavaScript/TypeScript parser

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P1-3.5 Implement fallback for unsupported languages

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P1-3.6 Add comprehensive tests

Verification: - [] AST parsing works for supported languages - [] Tests pass - [] Documentation updated

Task P1-4: Increase Test Coverage (Critical Packages)

Priority: P1 - HIGH **Estimated Effort:** 60 hours **Assigned To:** TBD (multiple developers)

Coverage Targets

Package	Current	Target	Tests Needed
internal/tools/browser	20.5%	80%	~50
internal/memory/providers	28.2%	80%	~80

Package	Current	Target	Tests Needed
internal/workflow/planmode	31.0%	80%	~30
internal/llm	42.0%	80%	~100
internal/server	56.4%	80%	~30

Subtasks

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P1-4.1 Browser package tests

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P1-4.2 Memory providers tests

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P1-4.3 Workflow planmode tests

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P1-4.4 LLM package tests

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P1-4.5 Server package tests

Verification: - [] Coverage reports show 80%+ - [] All tests pass - [] No flaky tests

Phase 2: MEDIUM Priority

Task P2-1: Application Test Coverage

Priority: P2 - MEDIUM **Estimated Effort:** 40 hours **Assigned To:** TBD

Subtasks

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P2-1.1 Desktop application tests (target 60%)

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P2-1.2 Terminal UI tests (target 60%)

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P2-1.3 Aurora OS tests (target 60%)

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P2-1.4 Harmony OS tests (target 60%)

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P2-1.5 Mobile binding tests (target 50%)

Task P2-2: Documentation Updates

Priority: P2 - MEDIUM **Estimated Effort:** 16 hours **Assigned To:** TBD

Subtasks

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P2-2.1 Update CLAUDE.md with accurate info

- Fix checkpoint interval documentation
- Document MCP transport status
- Update fallback documentation

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P2-2.2 Add missing READMEs

- cmd/config_test/README.md
- cmd/helix_config/README.md
- cmd/security_fix/README.md
- applications/android/README.md
- applications/ios/README.md

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P2-2.3 Update API documentation

- Document error responses
- Remove references to placeholder data

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P2-2.4 Update user manuals

- Add troubleshooting for database unavailable
- Document authentication requirements

Phase 3: LOW Priority

Task P3-1: Complete Remaining Coverage

Priority: P3 - LOW **Estimated Effort:** 40 hours **Assigned To:** TBD

Target: 100% coverage for all packages

Task P3-2: Enhancement Features

Priority: P3 - LOW **Estimated Effort:** 24 hours **Assigned To:** TBD

Subtasks

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- ☐ **P3-2.1** Multi-Edit Git integration
 - ☐ **P3-2.2** Multi-Edit rename operation
 - ☐ **P3-2.3** FAISS real compression
 - ☐ **P3-2.4** MCP stdio transport (if needed)
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Verification Checklist Template

Use this checklist for each completed task:

Task [ID] Verification

****Developer**:**
****Review Date**:**
****Reviewer**:**

Code Quality

- [] Code follows project style guide
- [] No new linting warnings
- [] No hardcoded values
- [] Error handling complete

Testing

- [] Unit tests written
- [] Integration tests written (if applicable)
- [] Tests pass locally
- [] Tests pass in CI
- [] Coverage target met

Documentation

- [] Code comments added
- [] README updated (if applicable)
- [] API docs updated (if applicable)

Security

- [] No security vulnerabilities introduced
- [] No mock data in production paths
- [] Authentication/authorization correct

Review

- [] Self-review completed

- [] Peer review completed
 - [] All review comments addressed
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How to Resume

This plan is designed to be pausable and resumable:

1. **Check Current Status:** Review the checkboxes above to see what's completed
2. **Pick Next Task:** Start with the lowest numbered incomplete task in the current phase
3. **Update Progress:** Check off subtasks as you complete them
4. **Verify:** Use the verification checklist before marking a task complete
5. **Update Status:** Update the overall status table at the top

Commands for Progress Check

```
# Check current coverage
cd /run/media/milosvasic/DATA4TB/Projects/HelixCode/HelixCode
go test -cover ./...

# Run specific package tests
go test -v -cover ./internal/server/...

# Generate coverage report
go test -coverprofile=coverage.out ./...
go tool cover -html=coverage.out -o coverage.html
```

Metrics to Track

Metric	Baseline	Current	Target
Total Coverage	47.5%	47.5%	100%
Critical Issues	15	15	0
High Issues	23	23	0
Mock Data Instances	12	12	0
Deprecated Dependencies	2	2	0

Document Version: 1.0 **Next Review Date:** TBD